DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022–23
HALF TEARLY EXAMINATION

CLASS: X | **FULL MARKS: 80**

SUBJECT: MATHEMATICS [SET B] | **TIME: $2\frac{1}{2}$ HOURS**

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

***The intended marks for questions or parts of questions are given in brackets* [ ].**

This paper contains 5 printed pages.

SECTION A
(*Attempt* ***all*** *questions from this section)*

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.) **[1×15]**

(i) The point (0, -5) is invariant under reflection in

(a) the origin (b) X-axis (c) Y- axis (d) Both X and Y axes

(ii) 14th term of the AP 7, 3, -1, -5, is
(a) 35 (b) 30 (c) -40 (d) -45

(iii) The order of the matrix $\begin{bmatrix} x + y & 2x \\ x - y & 2z - w \end{bmatrix}$

(a) 3×2 (b) 2×3 (c) 3×3 (d) 2×2

(iv) The remainder when $9x^2 - 6x + 2$ is divided by x -3, will be

(a) 60 (b) 65 (c)19/2 (d) 19/5

(v) If $2x + 1$ is factor of the polynomial $2x^2 + 5x + p$, then the value of p is:
(a) -3 (b) 2 (c) 3 (d) 7

(vi) If M (-3, 7) be the middle point of the line segment XY whose one end has the coordinates (0, 0), then the coordinates of the other end is:

(a) (-6, 7) (b) (-3, 7) (c) (-6, 14) (d) (3, -14)

(vii) If $23 - 4x \leq 5$, then what is the smallest value of x when x is a whole number?
(a)4 (b) 0 (c) 5 (d) -4

(viii) Nature of roots of the given equation $5x^2 -2x + \frac{1}{5} = 0$ will be

(a) equal and real (b) equal, real and irrational

(c) not equal but real and rational (d) imaginary

(ix) If 3, 12, x and 32 are in proportion, then the value of x is:

(a) 6 (b) 8 (c) 4 (d) 2

(x) Net tax paid by dealer to the Government is:
(a) Output Tax - Input Tax (b) Input Tax - Output Tax
(c) Input Tax (d) Output Tax

(xi) $\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ a kind of:

a) diagonal matrix b) identity matrix c) null matrix d) row matrix

(xii) A die is thrown once, then the probability of getting a number more than 4 is :

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{4}{6}$ (d) $\frac{1}{4}$

(xiii) Smita deposited ₹600 per month in a bank in cumulative account for 20 months. If she gets ₹ 13050 on maturity, then the amount of interest received is:
(a) ₹ 1040 (b) ₹ 2050 (c) ₹ 1050 (d) ₹ 1045

(xiv) If the slope of the line which passes through the points (2 , 5) and (k, 3) is 2, then the value of k is:

(a) -1 (b) -2 (c) 1 (d) 3

(xv) A vertical stick 20 m long casts a shadow 10 m long on the ground. At the same time, a tower casts a shadow 50 m long on the ground. The height of the tower is:
a) 100 m (b) 120 m (c) 25 m (d) 200 m

Question 2

a) Mohan deposits ₹ 800 per month in a cumulative deposit scheme for six years. Find the amount payable to him on maturity, if the rate of interest is 6% per annum. [4]

b) If b is the mean proportional between a and c, prove that:

$$\frac{a^2-b^2+c^2}{a^{-2}-b^{-2}+c^{-2}} = b^4$$ [4]

c) The 4th term of an A.P. is zero. Prove that the 25th term of the A.P. is three times its 11th term. [4]

Question 3

a) Solve the equation for x and write your answer correct to two decimal places:

$2x - \frac{1}{x} = 2$ [4]

b) Find the equation of the straight line which is parallel to the line 9x - 3y = 5 and passes through the point (–2, –5). [4]

c)Use graph paper for this question (Take 2 cm = 1 unit along both x and y axis). ABCD is quadrilateral whose vertices are (3,4) , (3,-4), (0,-1), (0,1).

i) Reflect quadrilateral ABCD under y axis and name it as A'B'CD.

ii) Write down the coordinates of A' and B'.

iii) Name two points which are invariant under the above reflection.

iv) Name the quadrilateral A'B'CD. [5]

SECTION B

(*Attempt **any four** questions from this section)*

Question 4

a) Two dice are thrown at a time. Find the probability of getting [3]

i)a doublet

ii)Sum of numbers appeared on dice is at least 10.

b) Find the values of x, y and z if $\begin{bmatrix} x+2 & 6 \\ 3 & 5z \end{bmatrix} = \begin{bmatrix} -5 & y^2+y \\ 3 & -20 \end{bmatrix}$ [3]

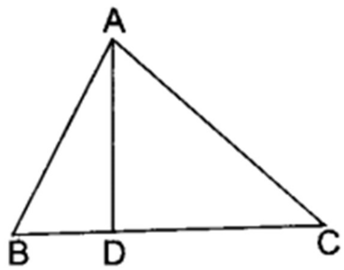
c) Mrs. Achariya buys goods worth ₹7500 from a store where she gets a rebate of 10% on ₹6000 as a member and a discount of ₹100 on the remaining. If the GST is charged at the rate of 18% find the total amount she has to pay for the goods. [4]

Question 5

a) When the two polynomials $x^3 - px^2 + x + 6$ and $2x^3 - x^2 - (p+3)x - 6$ are divided by (x-3), the remainder is same. Find the value of p. [3]

b) If 7 times the 7th term of an A.P. is equal to 11 times its 11th term, then find its 18th term. [3]

c) In triangle ABC, ∠BAC = 90° and AD ⊥ BC. Then show that BD.CD = AD² [4]



Question 6

a) If $x \in I$, then find the solution set of the inequation $1 < 3x + 5 \leq 11$ and plot it on the number line. [3]

b) If the points A (1, 2), O (0, 0) and C (a, b) are collinear, then find a relation between a and b. [3]

c) If $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$ and $A^2 - kA - 5I = 0$, then find the value of k. [4]

Question 7

a) Find the value of k for which the following equation has coincident roots.
$x^2 + 4kx + (k^2 - k + 2) = 0$ [3]

b) Find the lowest value of x: $\frac{x}{4} < \frac{5x-2}{3} - \frac{7x-3}{5}$; $x \in N$ [3]

c) If $x = \frac{\sqrt{a^2+b^2}+\sqrt{a^2-b^2}}{\sqrt{a^2+b^2}-\sqrt{a^2-b^2}}$, then show that $b^2 = \frac{2a^2x}{x^2+1}$ [4]

Question 8

a) Find the number to be subtracted from 8, 13, 16, 31 to make them proportional. [3]

b) Find the equation of a line that is parallel to the line $y = 3x - 5$ and whose y-intercept is (-1/5). [3]

c) Mayank covers a distance of 1 km from A to B. While returning, his speed was 3 km/hr more than the speed of first half of the journey. The total time taken for the journey is half an hr. Find the speed for the first half and second half of his journey. [4]

Question 9

a) Two coins are tossed together. Write the sample space. what is the probability of getting i) only Heads ii) at most one Head? [3]

b) If ABCD is parallelogram, P is a point on side BC and DP when produced meets AB produced at L, then prove that DP. BL = DC.PL [3]

c) Use the factor theorem to factorize the following expression:
$2x^3 - 5x^2 + x + 2$. [4]

Question 10

a) If P' (-4, -2) is the image of P under reflection in origin,
i) find the coordinates of P
ii) the coordinates of the image of P under the reflection in the line y = -2. **[3]**
b) How many terms of A.P. 20, 18, 16,…. are needed to give sum zero? **[3]**
c) If A = $\begin{bmatrix} 4 & 2 \\ 6 & -3 \end{bmatrix}$ **, B =** $\begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}$ **and C =** $\begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$**,** **[4]**
find i) $A^2 - A$ **ii) BC**